

Lighting Installation Guide

Version History

Version	Revised By	Changes	Date
1.0	Mark Veikos	Initial Version	4/12/2018
1.1	Mark Veikos	Ceiling Mounted Restroom Lights, Freezer & Cooler Lights	8/1/2018
1.2	Lee Kaufman	General Changes	10/5/2018
1.3	Raphaelle Canaan	Added SubEnergy Branding	7/21/21
1.4	Raphaelle Canaan	Ceiling Mounted Fixtures, Counter Menu Lights, Occupancy Sensors, Ballast Warning & Romex Wiring	4/13/22

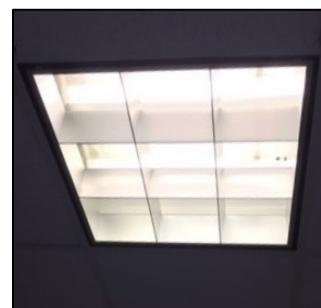
Last Saved: 5/4/2022 by Raphaelle Canaan

1. Overview

This guide will detail the steps to replace all existing lighting fixtures with LED equivalents.

All existing 2x2 and 2x4 fixtures will be replaced with both color and wattage adjustable LED fixtures. These fixtures will be “tuned” during the installation to the specific conditions of the store.

Other lighting lamps and fixtures will be replaced as outlined below, in colors of 3500K or 4000K.



Existing



New

Note: All fixtures containing ballasts must have ballast removed. Fixtures must also be converted to line-voltage with a warning sticker or sign attached.

Any NEC or local code wiring violations discovered during the DIRECT installation of the lights (improper wiring, exposed junction boxes, exposed bus bars at lighting panels, etc) should be corrected as part of the lighting installation. Overall code violations not directly related to the lighting installation should be reported.

Typically, Non LED exit signs or emergency only lights should ne replaced.

During the installation process, please consult with store manager if there are any question related to lighting levels and or what is being illuminated and if proposed solution is acceptable.

2. Install and Adjust Standard LED Fixtures

a. Install Recessed Fixtures

All existing 2x2 and 2x4 recessed fluorescent fixtures will be replaced with 2x2 LED fixtures. When replacing, keep in mind that these fixtures are typically 2-3x as bright as existing fluorescent 2x2s, and as bright or brighter than existing fluorescent 2x4s. The following steps should determine placement of 2x2s when replacing 2x4s:

- Maintain as much symmetry in store grid as possible.
- Ensure no fixture is touching another fixture in any direction, including diagonal.
- If a 2x4 is next to a wall, position the 2x2 on the side that is further from the wall.
- If still unsure on fixture placement, call the Budderfly representative that you have been in contact with for further instructions.

b. Ceiling Mounted Fixtures

Ceiling mounted fixtures will remain in place and should not be removed from the ceiling. There are exceptions for surface mounted lights in the “front of the house” where the customer deems the tube replacement inadequate.

These lights typically appear in restrooms and back areas. Instead of replacing the fixture, bulbs will be re-lamped with line voltage, ballast bypass 4000K LED tubes. Existing bulbs are typically 4ft T8 tubes, 2ft PL tubes, or 2ft U-tubes.



Fixtures with 2 existing lamps can be replaced with 2 - 14 Watts 4000K LED tube. Make sure fixtures have a 1 to 1 replacement for each lamp to prevent dark spots within the fixtures. For other fixtures replace 1 for 1.

For front of the house surface mounted fixtures that are deemed not to be a good “aesthetic” look to re-lamp, 2x2 and 2x4 fixtures are available with surface mount kits (pls check with your supervisor). PLEASE NOTIFY THE CUSTOMER THAT WE CAN NOT GUARANTEE THERE WILL BE AREAS THAT NEED TO BE REPAIRED OR REPAINTED. THIS WORK IS NOT COVERED BY BUDDERFLY.

Note: All fixtures containing ballasts must have ballast removed. Fixtures must also be converted to line-voltage with a warning sticker or sign attached.

c. Emergency Lighting / Exit Signage

In stores where, existing drop-in fixtures with battery-backup systems serve as the emergency lighting, these fixtures will be replaced with standard flat panel fixtures. A battery-backup emergency light should then be installed per Codes.



In the event of a non-LED exit sign or non LED exit sign with emergency lighting, replace with LED Exit Sign with or without emergency lights.

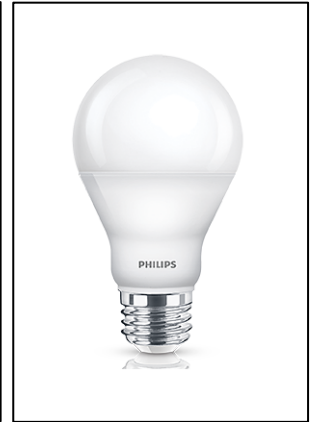
Existing LED exit signs with or without emergency lighting should be left as is.

3. Replace Other Existing Lighting

a. Dining Pendant Lights

Pendant lights hang down from the ceiling and contain screw-in A19 bulbs and other types of screw-in bulbs. Currently, the pendants may have either incandescent, compact fluorescent, or LED bulbs. If the bulbs are all incandescent or CFLs, replace entirely with LEDs.

If LEDs are already installed and bulbs are less than 8W each, keep in place. Otherwise, if existing LEDs are greater than 8W, replace with 40-Watt equivalent Soft White / Warm White LED Bulbs (typically ~ 5W).



b. Dining Track Lights

Some shops will have track lights installed around the edges of the shop to highlight wall art or provide decorative lighting. These are typically halogen PAR30 or PAR38 lights, and should be replaced with 10W or 12W LEDs.

c. Exterior Lights (standalone shops only)

In locations where the store is responsible for exterior lighting (not in strip malls or combined buildings), exterior lights should be replaced with LEDs.

Multiple types of exterior lights may be found at each shop, including spot and floodlights, downlights, and parking lot lights. Replace these with LED bulbs / fixtures of the equivalent wattage.

d. Counter Can Lights

Some shop may have recessed can lights throughout the shop or over the counter area, potentially housed in wood paneling. Take the following action depending on the existing light type:

- a. Existing LEDs – should be replaced to ensure light color match
- b. Existing CFL screw-in (Type A bulb) – replace with 40W equivalent (~5W) LED Bright White (3500K) screw-in bulb
- c. Existing CFL 4-pin bulb (G24q) – replace with 11W 4000K LED G24q plug-and-play bulb – ballast bypass whenever possible
- d. Existing low voltage MR16 – replace with equivalent wattage and voltage MR16 LED
- e. Other non-LED existing bulb/lamp – replace with equivalent wattage LED



e. Counter Menu Lights

Counter menu lights are typically 2, 3 or 4 foot T8 or T5 fluorescent bulbs and will be replaced with the equivalent line voltage LED tubes.

f. Emergency Lights

Appropriately address emergency and exit lighting with LED exit signs where applicable. Ensure compliance with all regional codes and safety standards. [More details to follow.]



g. Walk-in Freezer and Cooler Lights

Walk-in freezer and cooler typically contain “jelly jar” or surface mounted fluorescent strip fixtures. For “jelly jars”, replace bulb with 60W equivalent LED (~10W) medium-base bulbs with a color of 5000K (daylight), if they do not already contain LEDs. For strip fixtures, replace fluorescent tubes with 5,000K direct wire LED tube, bypassing the ballast.

4. Occupancy Sensors

Occupancy sensors should be installed in all “small rooms” (restrooms, office, utility closets, etc.). Occupancy sensors should not be installed in bathrooms where partitions (toilet or urinal) are present as they could prevent motion sensor from activating during movement. Light switches in the above areas should be replaced with occupancy sensors.



For fan/light combinations, the bulb in the light should be replaced, we do not replace restroom fans as part of our program

For the occupancy sensor to function with a fan/light combination, a neutral conductor must be present at the switch.

If the current light switch controls both fans and lighting and does not have a neutral, abort the occupancy sensor installation.

5. Post-Install

We will require you to fill out a post-install survey once a project is complete. Please keep this installation document for reference and take 5-10 pictures of the shop and ceiling post-install.

5. Romex Wiring

Per the most recent NEC (2020 NFPA 70), Romex, or type NM, wiring is generally not permitted in commercial applications. There are however some local jurisdictions that may permit Romex wiring or may consider existing Romex wiring to be “grandfathered”. It is up to the local Electrician to use his or her knowledge and

advise if the existing Romex wiring may be re-used. Under no circumstances should the lighting fixtures be replaced if Romex wiring is not permitted by the authority having jurisdiction (AHJ).

6. Miscellaneous Items

- Budderfly requires proof of licensure and insurance before working with any contractor. We will need a completed W-9 form from all contractors.
- All necessary permits in the jurisdiction need to be secured and approved before starting work.
- Follow all local codes during installation. This includes earthquake certification and stickers when fixtures or bulbs are powered by line voltage.
- Please ensure adequate protection for customers, workers, and food products during the installation process.
- We require proper disposal of all existing fixtures and lamps. Please retain all supporting documentation.